*Power Transformer Differential Relay Protection Using MATLAB/Simulink*

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# Abstract

This project presents a MATLAB/Simulink model for a power transformer differential protection relay. The main purpose of the project is to protect a three-phase power transformer against internal faults by comparing the currents on both sides of the transformer. The simulated transformer is rated at 200 MVA, 11 kV / 33 kV, 50 Hz, with grounded-wye connections on both sides.

The relay algorithm is designed using a MATLAB Function block. It calculates the differential current and the restraint current then decides whether the fault is inside or outside the protected zone. The relay also includes practical features such as percentage differential protection, a dual-slope characteristic, second harmonic blocking for inrush current, fifth harmonic restraint for overexcitation high set differential tripping, trip latching and circuit breaker control.

The system was tested under four scenarios: normal operation, internal fault, external fault and transformer energization with inrush current. The simulation results showed that the relay did not trip during normal operation or external faults. it successfully operated during an internal fault and sent an opening command to the circuit break (CB).

# 1. Introduction

Power transformers are important components in electrical power systems because they transfer power between different voltage levels. They are also expensive and critical for system operation, so protecting them from faults is very important. A transformer fault can cause serious equipment damage and may affect the stability and continuity of the power system [1].

One of the most common protection methods used for power transformers is differential protection. The main idea is to compare the current entering the protected zone with the current leaving it. Under normal operating conditions, these currents should be almost balanced after considering the transformer ratio and current direction. In this case the differential current remains small and the relay should not trip [1, 2].

When an internal fault occurs inside the protected zone the current balance between the two sides of the transformer is disturbed. As a result, the differential current increases. If this current becomes higher than the relay setting, the relay sends a trip signal to open the circuit breaker (CB) and isolate the faulted transformer. This makes differential protection a fast and selective method for transformer internal fault detection [2, 3].

Transformer differential protection is not always straightforward. Some conditions may produce high currents even when there is no internal fault. A common example is magnetizing inrush current which appears when the transformer is energized. This current can be large and may cause false tripping if the relay does not recognize it correctly. For this reason, second harmonic blocking or restraint is commonly used since transformer energization can produce a large inrush current with a noticeable second harmonic component [3, 4].

Another condition that should be considered is transformer overexcitation. Harmonic restraint and blocking methods are commonly used in transformer differential relays to improve relay security and reduce the possibility of unwanted operation during non-fault conditions [4, 5].

# 2. System Description

## 2.1 Complete System Model

Figure1.below shows the complete MATLAB/Simulink model of the proposed transformer differential protection system. The model consists of a three-phase source, CB, a three-phase two-winding transformer, current measurement blocks, fault blocks, and a three-phase load. The transformer is rated at 200 MVA, 11 kV / 33 kV, 50 Hz with grounded-wye connections on both sides.

Current measurements from the primary and secondary sides are sent to a relay subsystem implemented using a MATLAB Function block. The relay calculates the differential current, restraint current, and harmonic content to detect internal faults and distinguish them from external faults and transformer inrush conditions. The protection algorithm also includes second harmonic blocking to improve relay security during transformer energization and prevent unwanted tripping caused by magnetizing inrush current.

In addition, several scopes and display blocks are connected throughout the model to monitor the measured currents, relay outputs, harmonic behavior and CB status during the simulation process. The relay generates a trip signal and a CB command when an internal fault is detected, allowing the transformer to be isolated from the power system quickly.

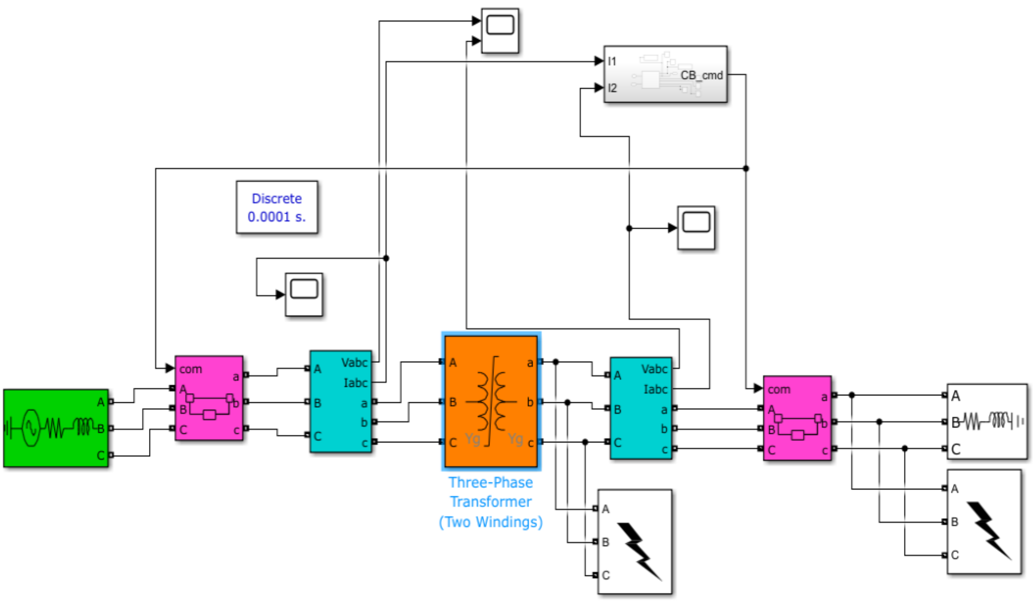



Figure 1. Power transformer differential protection system.

## 2.2 Relay Subsystem

Figure 2. shows the internal structure of the differential relay subsystem. The relay receives two three-phase current signals, $I_1$ and $I_2$ from the current measurement blocks installed on both sides of the transformer. These signals are used as the main inputs to the MATLAB Function block where the protection algorithm is implemented.

Inside the relay algorithm the measured currents are processed to calculate the differential current ($I_d$), restrain current ($I_r$) and second harmonic ratio ($H_2$). The differential current is used to detect internal faults while the restraint current improves relay stability during high-current conditions and external faults. The second harmonic ratio is used to identify transformer inrush current and prevent false tripping during transformer energization.

The relay produces several output signals for monitoring and control. The Trip signal indicates whether the relay has detected an internal fault. The Circuit Breaker command (CB_cmd) signal is sent to the CB where a value of 1 keeps the breaker closed and a value of 0 opens the breaker. Other outputs such as Id_display, Ir_display, H2_display, Inrush Block and inrush current are connected to scopes and displays to monitor the relay behavior during different test cases.

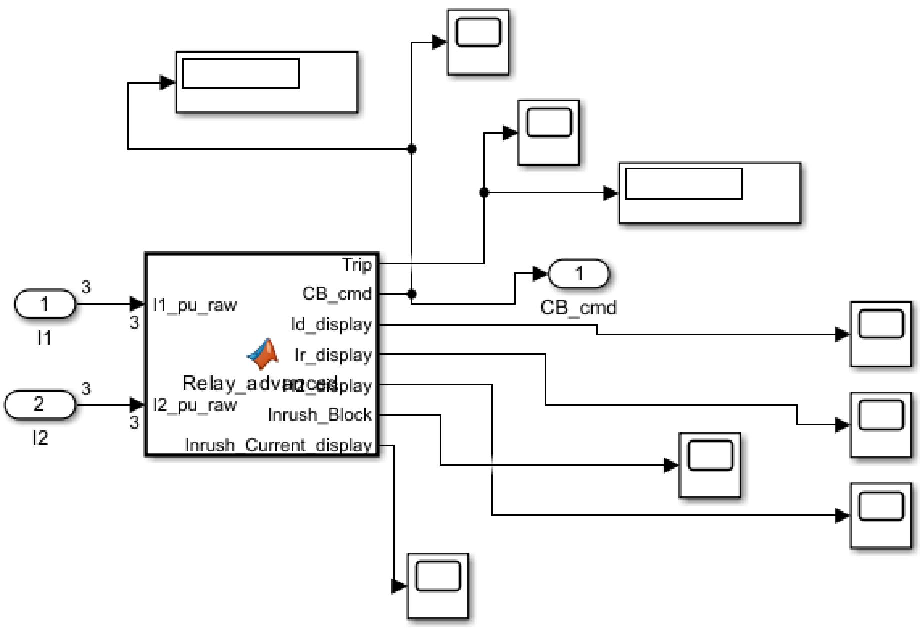



Figure 2. Differential relay subsystem.

# 3. Protection Methodology

## 3.1 Differential Current Calculation

The differential current is the main operating quantity used in transformer differential protection. It represents the difference between the currents entering and leaving the transformer protection zone. Under normal operating conditions, the primary and secondary currents are approximately equal after ratio compensation resulting in a very small differential current. During an internal fault inside the transformer zone the difference between the two currents increases significantly causing the relay to operate [1, 2].

$$I_d = |I_1 - I_2| \qquad (1)$$

Where:

$I_d$: is the differential current.

$I_1$: is the primary-side current in per-unit.

$I_2$: is the secondary-side current in per-unit.

In this project both $I_1$ and $I_2$ are measured in per-unit (pu) values using the current measurement blocks in Simulink. This allows the relay to compare both transformer side currents on a common base before calculating the differential current.

## 3.2 Restraint Current Calculation

The restraint current is used to improve the stability and security of the differential protection relay during external faults and normal operating conditions. It prevents unnecessary relay operation caused by measurement errors, current transformer mismatch, or transient conditions. The restraint current used in this project is calculated as the average magnitude of the primary-side and secondary-side currents [1, 2].

$$I_r = \frac{|I_1| + |I_2|}{2} \qquad (2)$$

Where:

$I_r$: is the restraint current.

$I_1$: is the primary-side current in per-unit.

$I_2$: is the secondary-side current in per-unit.

## 3.3 Harmonic Blocking Technique

Transformer energization produces a large magnetizing inrush current that may cause the differential current to increase temporarily even when no internal fault exists. To avoid false relay operation during this condition, second harmonic blocking is included in the protection algorithm [3, 4].

$$H_2 = \frac{I_{2nd}}{I_{1st}} \qquad (3)$$

Where:

$H_2$: is the second harmonic ratio.

$I_{2nd}$: is the magnitude of the second harmonic component.

$1st$: is the magnitude of the fundamental current component.

When the second harmonic content exceeds the specified threshold, the relay identifies the condition as transformer inrush current and blocks the trip signal to prevent unnecessary breaker operation.

## 3.4 Trip Decision Logic

The trip decision logic combines the differential current calculation, restraint current supervision and harmonic blocking technique to determine the operating state of the relay. Under normal operating conditions and external faults, the relay remains stable because the differential current remains below the operating threshold. During internal faults the differential current increases significantly while the harmonic content remains low causing the relay to generate a trip signal [1, 3].

$$I_d > I_{pickup} \qquad (4)$$

The relay continuously monitors the system currents during the simulation process. Once the differential current exceeds the pickup setting and harmonic blocking is inactive, the relay sends a trip command to the CB to isolate the transformer from the power system.

# 4. Differential Relay Operating Principle

The transformer differential relay operates by comparing the currents entering and leaving the protected transformer zone. In this project these currents are measured from the primary and secondary sides of the transformer using current measurement blocks in Simulink. Under normal operating state and external faults, the two measured currents remain approximately balanced after ratio compensation. The differential current remains very small and the relay does not operate [1, 2].

During an internal fault inside the transformer protection zone, the balance between side currents is disturbed. As a result, the calculated side and secondary the primary differential current increases above the relay operating threshold. In the developed Simulink model this condition causes the relay to generate a trip signal and changes the CB command from 1 to 0 which opens the breaker and isolates the transformer from the power system [1, 3].

To improve relay security during transformer energization second harmonic blocking is included in the protection algorithm. This feature is important because magnetizing inrush current may produce a high differential current even when there is no internal fault. Since inrush current usually contains a noticeable second harmonic component, the relay uses the second harmonic ratio to block unnecessary tripping during transformer energization [3, 4].

# 5. Relay Algorithm

The relay algorithm was implemented using a MATLAB Function block in Simulink. It receives three-phase current signals from both sides of the transformer and processes them to detect internal faults. The algorithm calculates the differential current restraint current and harmonic components then uses these values to decide whether the relay should trip or remain stable.

## 5.1 Relay Operating Steps

The operating sequence of the relay algorithm is as follows:

1. Measure the currents
The relay receives $I_1$ and $I_2$, which represent the primary-side and secondary-side three-phase currents in per-unit.

2. Store the samples
The current samples are stored in a one-cycle buffer to allow proper signal processing.

3. Extract current components
A Discrete Fourier Transform (DFT) is used to extract the fundamental, second harmonic, and fifth harmonic components.

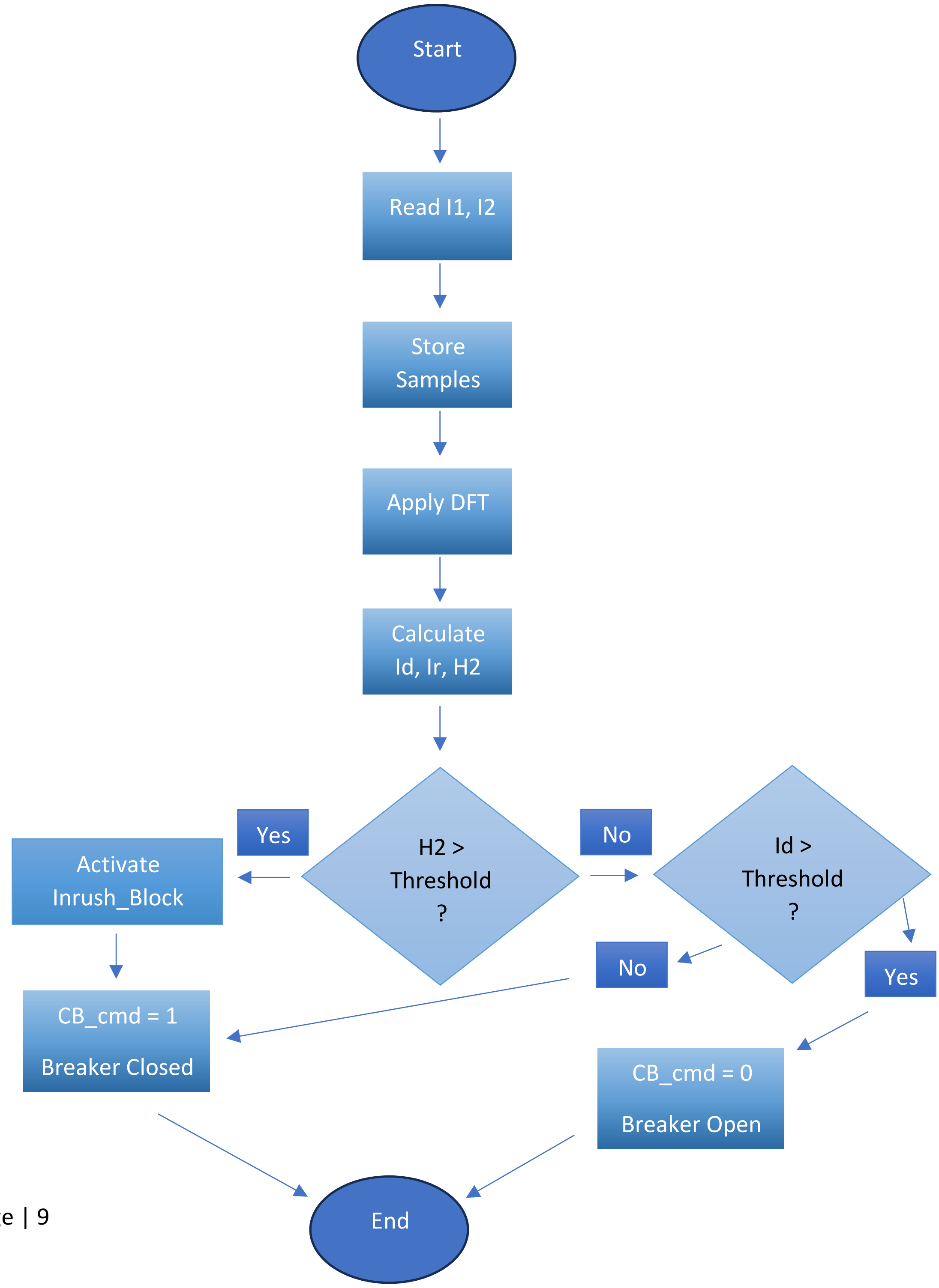
4. Calculate protection quantities
The relay calculates the differential current $I_d$, restraint current $I_r$ and second harmonic ratio $H_2$.

5. Check the inrush condition
If the second harmonic ratio exceeds the selected threshold, the relay activates the inrush blocking signal.

6. Make the trip decision
If $I_d$ is greater than the operating threshold and inrush blocking is inactive, the relay generates a trip signal.

7. Send the breaker command
When the relay trips the CB_cmd changes from 1 to 0 causing the CB to open and isolate the transformer.

## 5.2 Relay Algorithm Flowchart



# 6. Scenarios and Results

The relay was tested under four conditions: normal operation, internal fault, external fault and transformer inrush current. For the internal and external fault tests, a three-phase-to-ground fault was applied by selecting phases A, B, C, and Ground in the Three-Phase Fault block. The main monitored signals were $I_1$, $I_2$, $I_d$, Trip, Circuit Breaker command (CB_cmd), $H_2$, and Inrush_Block.

In this model, Trip = 1 means that the relay has detected an internal fault. The breaker control signal is CB_cmd, where CB_cmd = 1 keeps the breaker closed and CB_cmd = 0 opens the breaker. when the relay trips, Trip becomes 1 and CB_cmd becomes 0.

## 6.1 Normal Operation

During normal operation no fault was applied. The primary and secondary currents were almost balanced, so the differential current is very small. Trip stayed 0 and CB_cmd stayed 1, which means the circuit breaker remained closed.

Figure 3. below shows the primary-side current ($I_1$) waveform during normal operation. The current remains stable and balanced because no fault is applied.

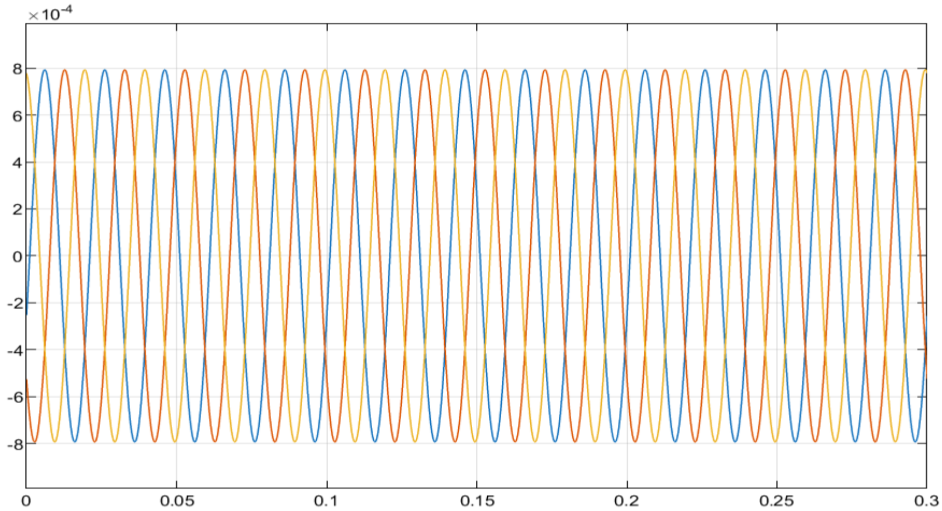



Figure 3. Primary-Side Current $I_1$ during Normal Operation.

Figure 4. shows the secondary-side ($I_2$) current waveform during normal operation. The waveform is almost balanced with the primary-side current, which indicates normal transformer operation.

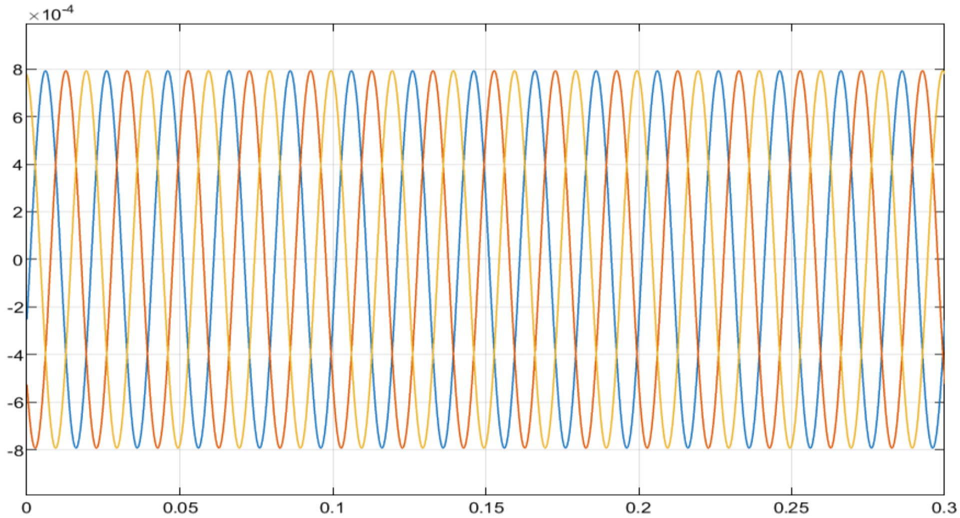



Figure 4. Secondary-Side Current $I_2$ during Normal Operation.

Figure 5. below shows that the trip signal remains at 0 during normal operation. This means that the relay does not issue a trip command because no internal fault is detected.

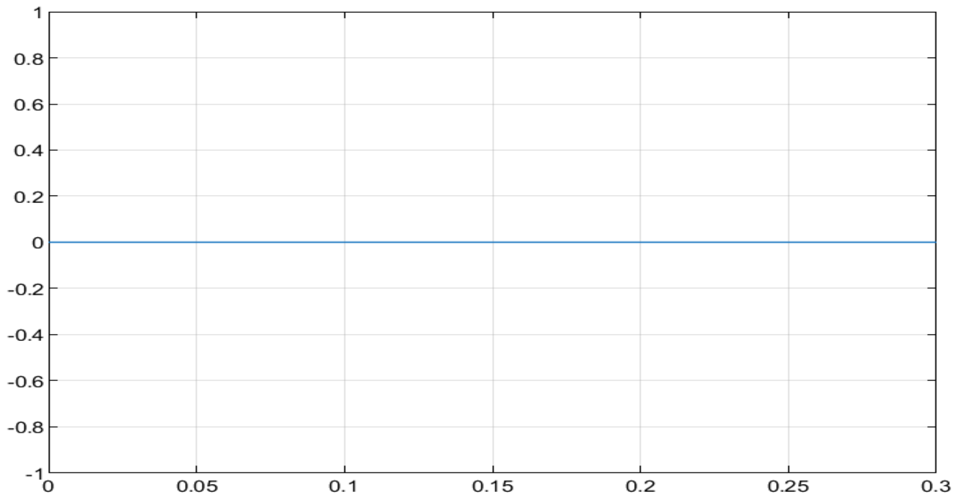



Figure 5. Trip Signal during Normal Operation.

Figure 6. shows that CB_cmd remains at 1 during normal operation. This means that the CB stays closed.

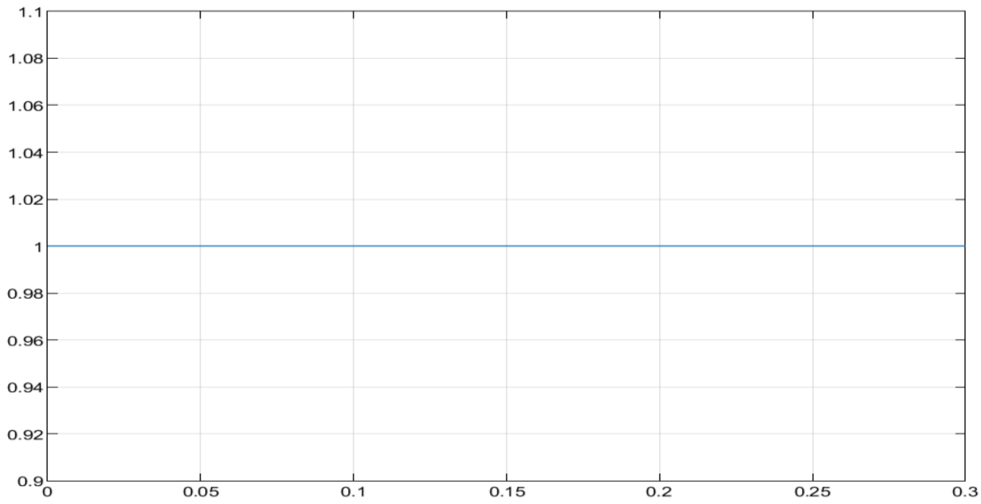



Figure 6. CB_cmd during Normal Operation.

## 6.2 Internal Fault

In this scenario an internal fault was applied inside the transformer protection zone at $t = 0.2$s. After the fault occurred the balance between the primary-side and secondary-side currents was disturbed causing the differential current to increase. As a result, the relay detected an internal fault trip changed from 0 to 1, and CB_cmd changed from 1 to 0 which caused the CB to open and isolate the transformer.

Figure 7. Below shows the primary-side ($I_1$) current during the internal fault test. A clear transient appears at the fault instant around 0.2 s. After the relay operates and the CB opens the current decreases to zero confirming that the faulted transformer is isolated.

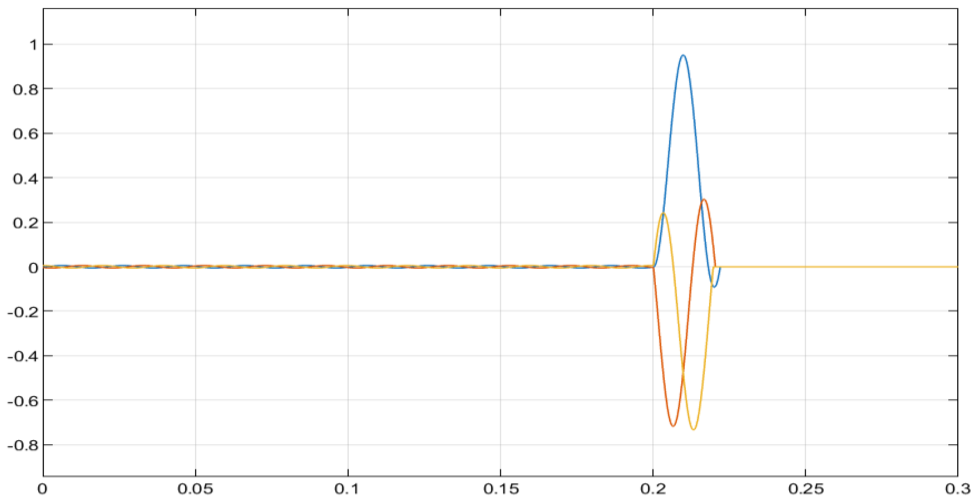



Figure 7. Primary-Side Current $I_1$during Internal Fault.

Figure 8. shows the secondary-side ($I_2$) current during the internal fault test. Before the fault, the current waveform is stable and balanced. At about 0.2 s, the fault causes a disturbance in the current waveform. After the relay operates and the circuit breaker opens, the current drops to nearly zero, which confirms that the transformer is disconnected from the system.

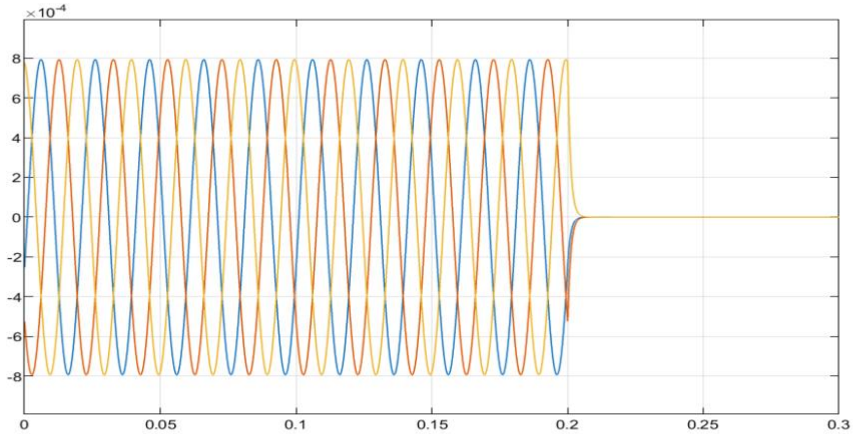



Figure 8. Secondary-Side Current $I_2$during Internal Fault.

Figure 9. Below shows the differential current ($I_d$) response during the internal fault test. After the fault is applied at about 0.2s the differential current increases clearly above its normal value. This confirms that the relay detects an unbalance between the primary-side ($I_1$) and secondary-side ($I_2$) currents.

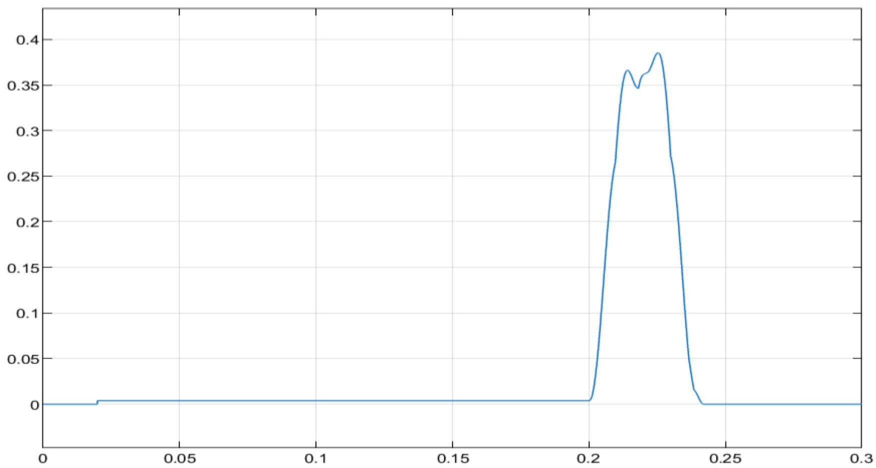



Figure 9. Differential Current $I_d$during Internal Fault.

Figure 10. Below shows that the trip signal changes from 0 to 1 after the internal fault is detected. This means that the relay correctly identifies the fault as an internal fault and issues a trip command.

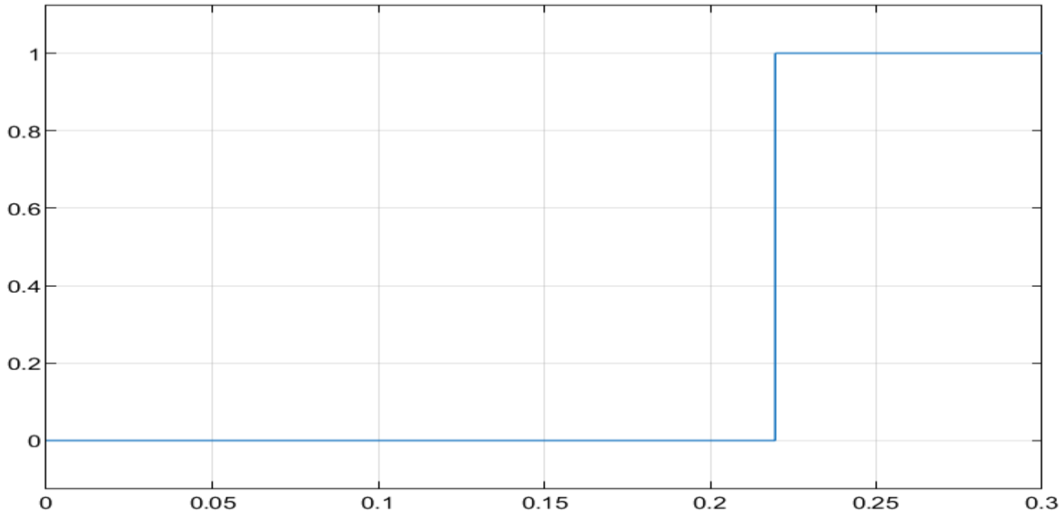



Figure 10. Trip Signal during Internal Fault.

Figure 11. Below shows that CB_cmd changes from 1 to 0 after the relay trip. Since the circuit breaker opens when CB_cmd = 0 this confirms that the transformer is isolated from the system after the internal fault.

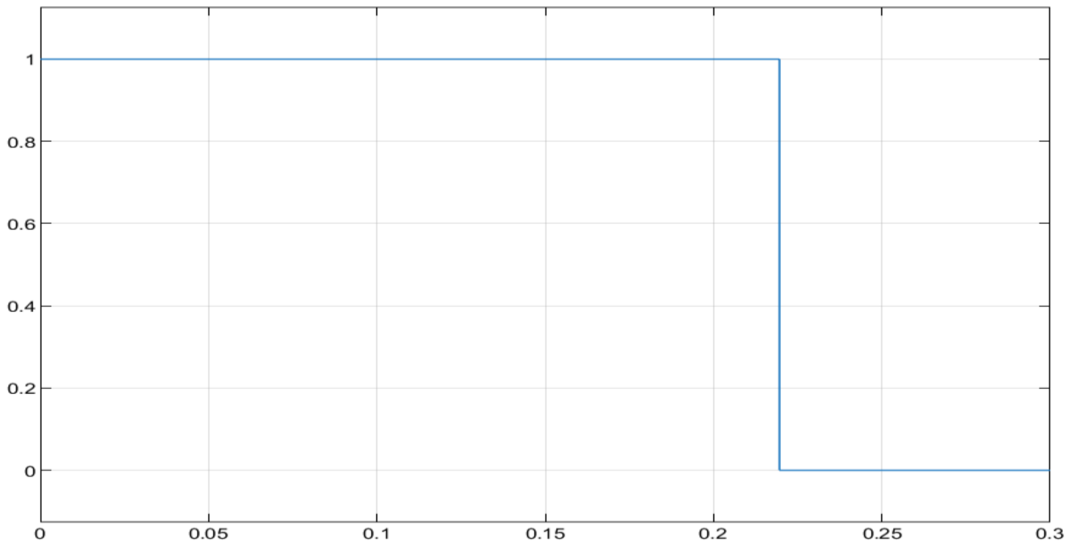



Figure 11. CB_cmd during Internal Fault.

## 6.3 External Fault

In this scenario the fault was applied outside the transformer protection zone. During the external fault, both the primary-side ($I_1$) and secondary-side ($I_2$) currents increased because the fault current passed through the transformer. The currents remained almost balanced from the relay point of view, so the differential current stayed small. As a result, trip remained 0 and CB_cmd remained 1 which means the circuit breaker stayed closed.

Figure 12. shows the primary-side current $I_1$during the external fault test. At 0.2 s $I_1$increases due to the external fault condition. The waveform becomes disturbed and its magnitude rises which indicates that the transformer is supplying fault current. Since the fault is outside the protected zone this condition does not cause relay tripping.

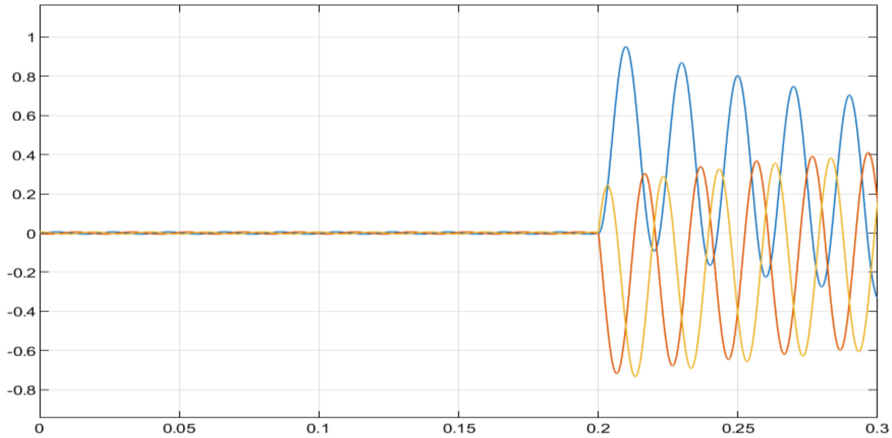



Figure 12. Primary-Side Current $I_1$during External Fault

Figure 13. shows the secondary-side current $I_2$during the external fault test. The waveform of $I_2$ also increases after the fault is applied like the primary-side current $I_1$. This behavior confirms that the current is passing through the transformer toward the external fault rather than being caused by an internal transformer fault.

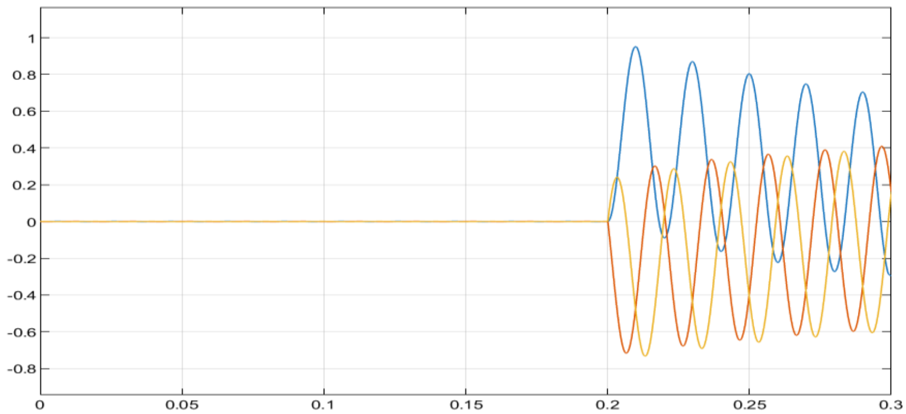



Figure 13. Secondary-Side Current $I_2$during External Fault

Figure 14. shows that the differential current $I_d$ remains very small during the external fault. Although the primary-side current $I_1$ and the secondary-side current $I_2$ increase the difference between them remains low. This confirms that the relay correctly identifies the fault as external and remains stable.

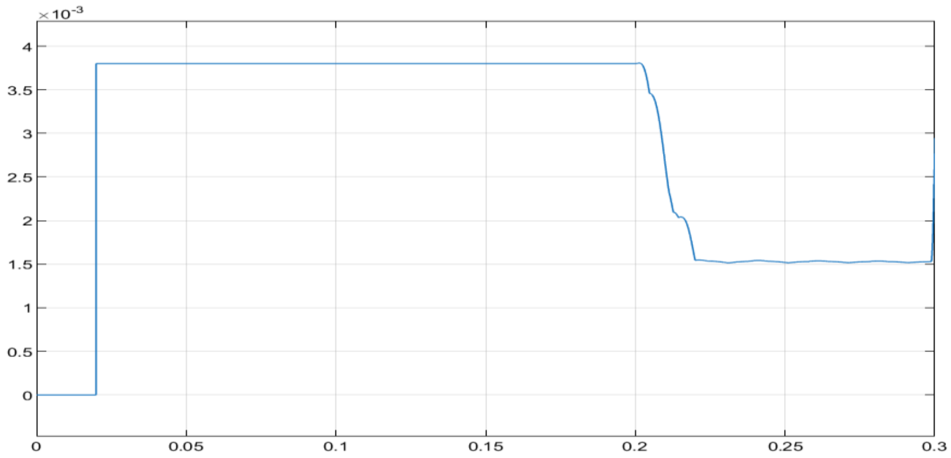



Figure 14. Differential Current $I_d$during External Fault.

Figure 15. below shows that the Trip signal remains at **0** during the external fault. This means that the relay does not issue a trip command because the fault is outside the transformer protection zone.

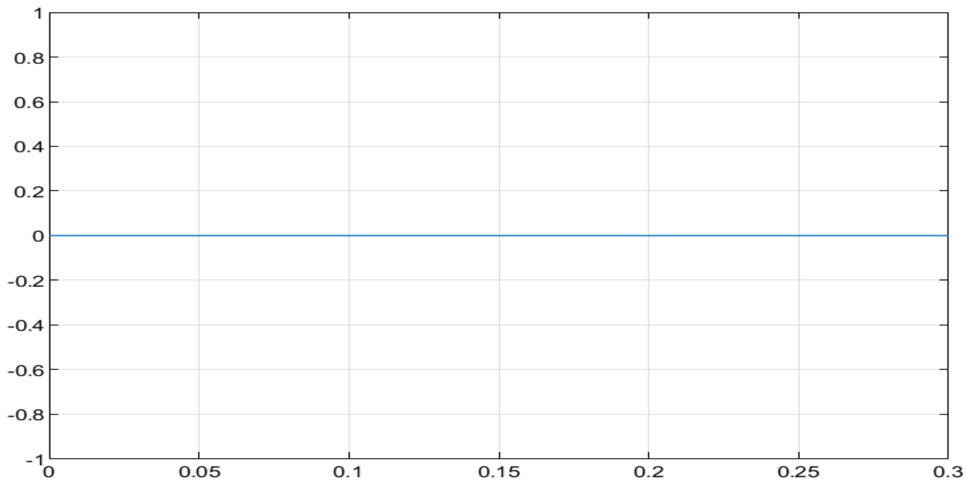



Figure 15. Trip Signal during External Fault.

Figure 16. below shows that CB_cmd remains at 1 during the external fault. Since the CB stays closed when CB_cmd = 1 this confirms that the transformer is not disconnected for an external fault. This result shows that the relay operates selectively and avoids unnecessary breaker operation.

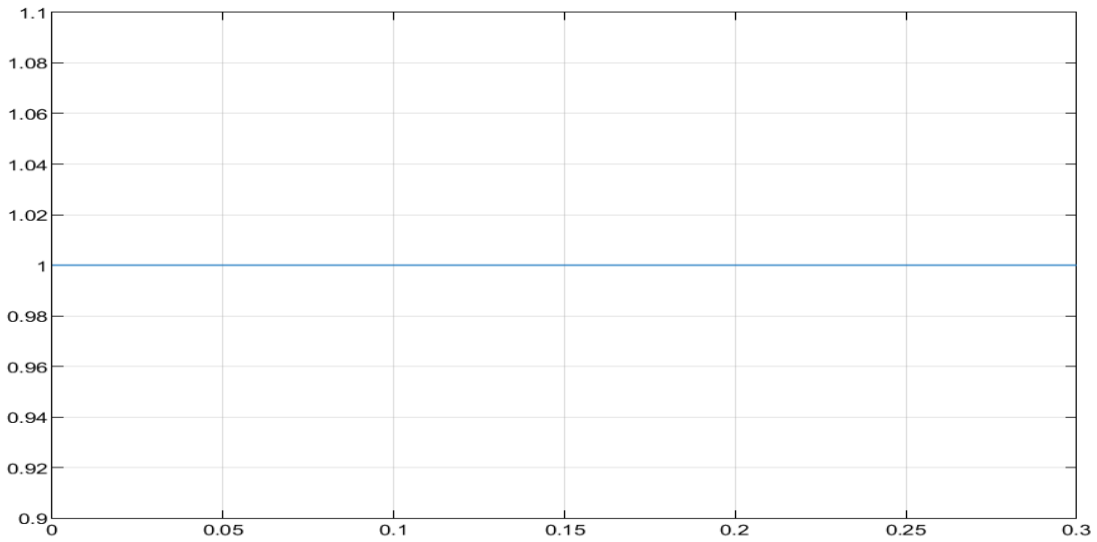



Figure 16. CB_cmd during External Fault.

## 6.4 Inrush Current Blocking

In this scenario, all fault blocks were disabled by setting the fault switching time to 10 which is outside the simulation time range. This ensures that no internal or externals fault occurs during the test. To create the transformer energization condition, the source-side circuit breaker was initially open and then closed at approximately 0.1s. Therefore, the transformer was suddenly connected to the supply at this time.

Transformer saturation was enabled in the model to produce magnetizing inrush current during energization. The relay response in this test is related only to transformer energization, not to any fault condition. The purpose of this test is to verify that the relay can recognize the inrush condition using the second harmonic content and block false tripping, while keeping Trip = 0 and CB_cmd = 1.

Figure 17. shows the primary-side current ($I_1$) during transformer energization with the fault blocks disabled. Before 0.1 s the current is zero because the transformer is disconnected from the source. At approximately 0.1s the source-side circuit breaker closes, and a magnetizing inrush current appears with a peaked and asymmetric waveform. This behavior is expected during transformer energization due to core saturation.

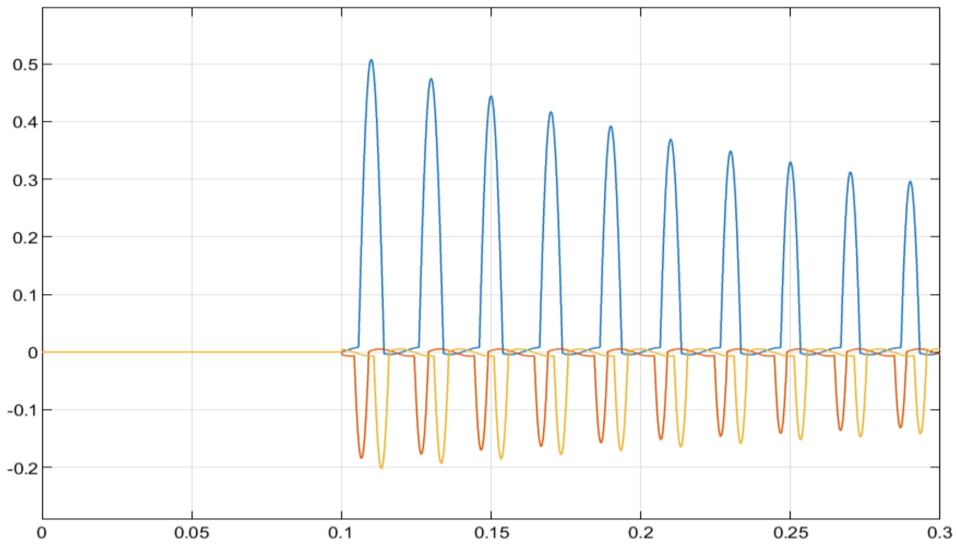



Figure 17. Primary-Side Current $I_1$during Transformer Energization.

Figure 18. shows the secondary-side current $I_2$during transformer energization. Before 0.1 s the current was nearly zero because the transformer was disconnected from the source. At approximately 0.1s the source-side circuit breaker closes and the transformer is energized. After that, the secondary-side current appears and reaches a balanced sinusoidal waveform.

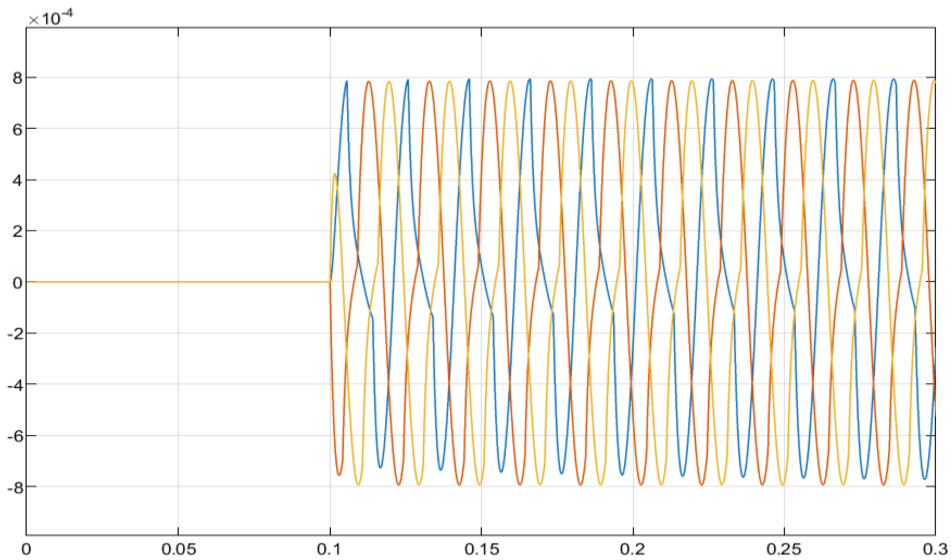



Figure 18. Secondary-Side Current $I_2$during Transformer Energization.

Figure 19. present the second harmonic ratio $H_2$ during transformer energization. At approximately 0.1s the transformer is energized and $H_2$ rises clearly. Since the second harmonic ratio is higher than the selected blocking threshold the relay identifies this condition as transformer inrush current. This prevents false tripping during transformer energization.

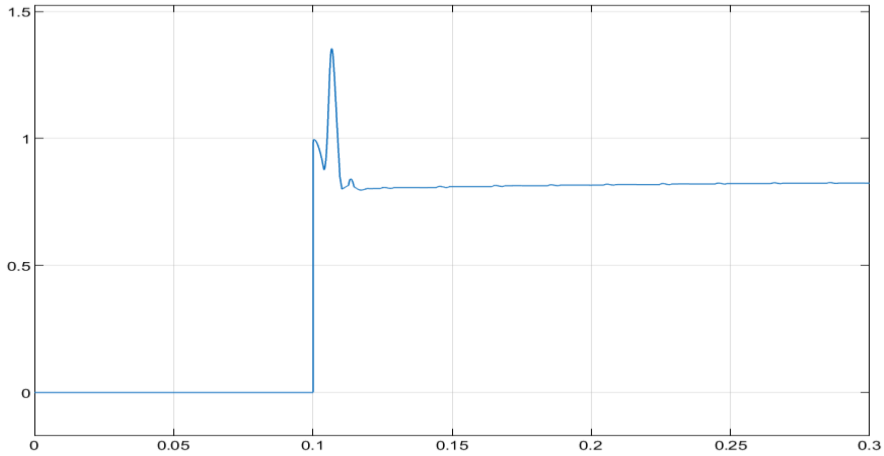



Figure 19. Second Harmonic Ratio $H_2$during Transformer Energization.

Figure 20. shows the Inrush Block signal during transformer energization. At approximately 0.1s the signal changes from 0 to 1, which means that the relay detects the inrush condition and activates the blocking function. This prevents the relay from issuing a false trip during transformer energization.

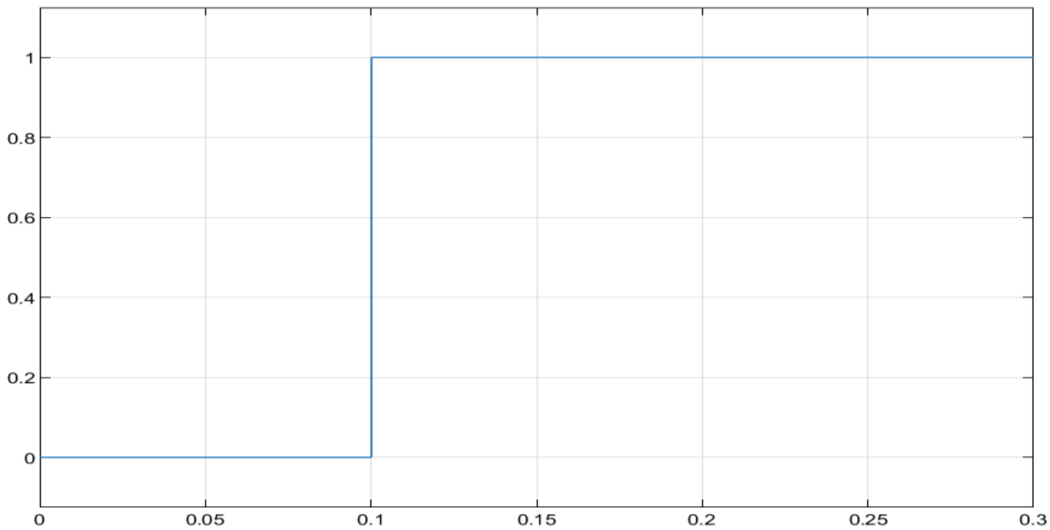



Figure 20. Inrush Blocking Signal during Transformer Energization.

Figure 21. present the Trip signal during transformer energization. Although the second harmonic ratio increases and the inrush blocking signal becomes active the Trip signal remains at 0. This confirms that the relay does not operate during the inrush condition.

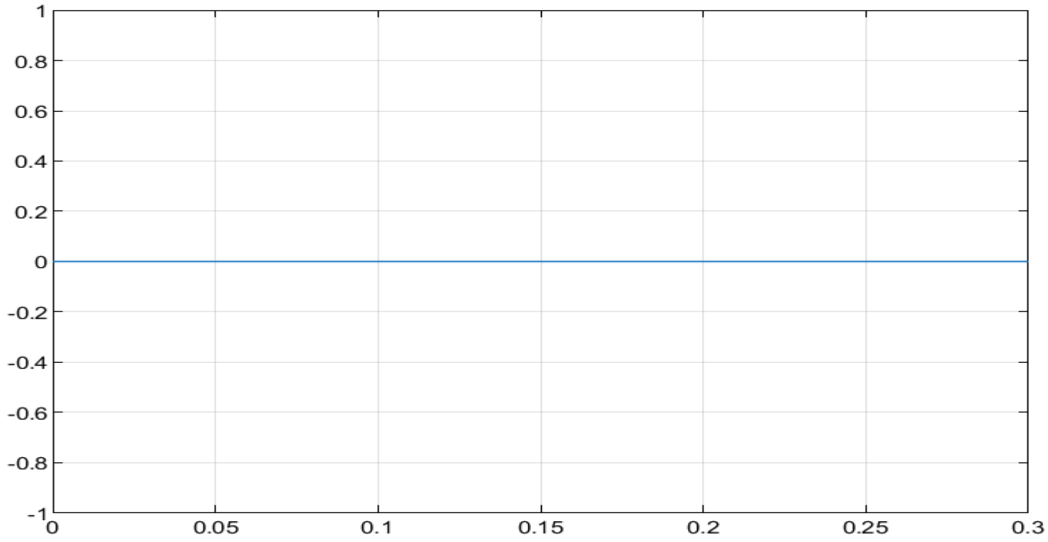



Figure 21. Trip Signal during Transformer Energization.

Figure 22. shows the circuit CB_cmd during transformer energization. The signal remains at 1 which means that the circuit breaker stays closed. This confirms that the relay remains stable and avoids unnecessary CB operation during transformer energization.

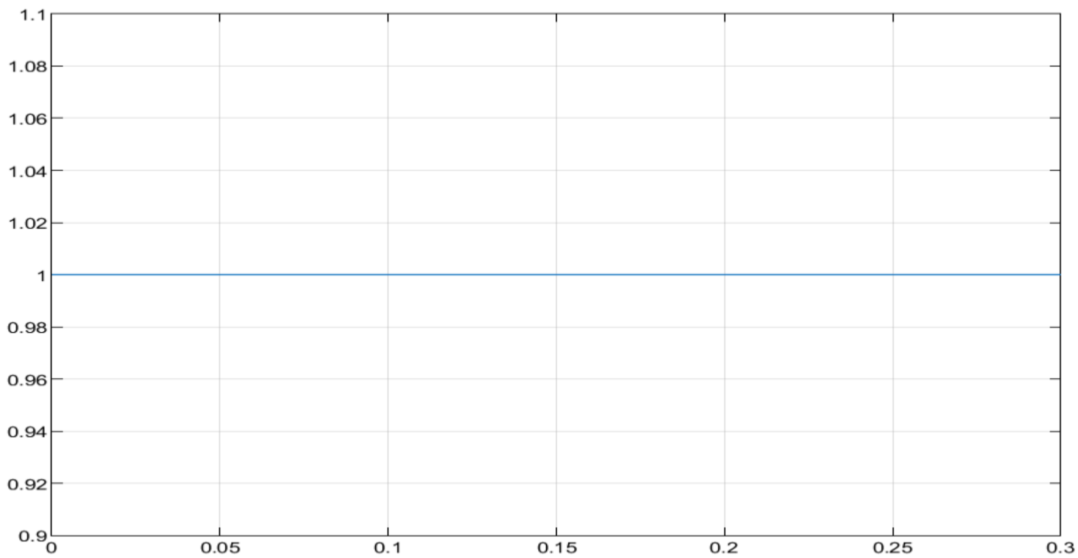



Figure 22. CB_cmd during Transformer Energization.

Figure 23. below shows the inrush current indicator during transformer energization. Before 0.1 s, the current is nearly zero because the transformer is disconnected from the source. At approximately 0.1 s, the transformer is energized and a transient magnetizing inrush current appears. The current gradually decreases as the transformer approaches steady-state operation.

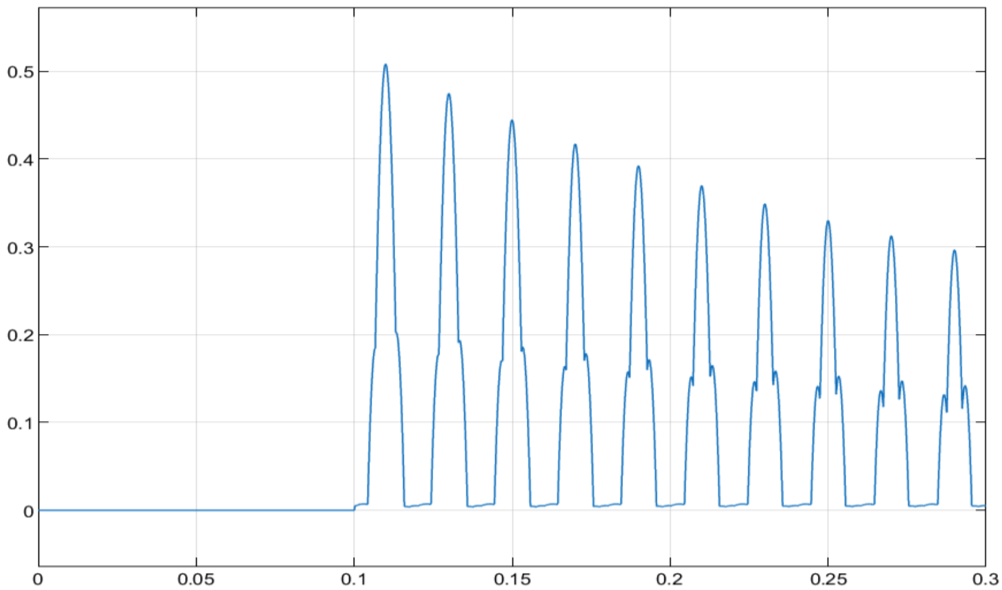



Figure 23. Inrush Current Indicator during Transformer Energization.

## 6.5 Single-Phase-to-Ground Internal Fault

In this test case, a single-phase-to-ground internal fault was applied inside the transformer protection zone. The fault was created by selecting Phase A and Ground only in the Three-Phase Fault block.

Figure 24. shows the primary-side current $I_1$ during the single-phase-to-ground internal fault. The waveform becomes disturbed mainly around the faulted phase after the fault is applied at 0.2s. This indicates that the system is affected by an unbalanced internal fault.

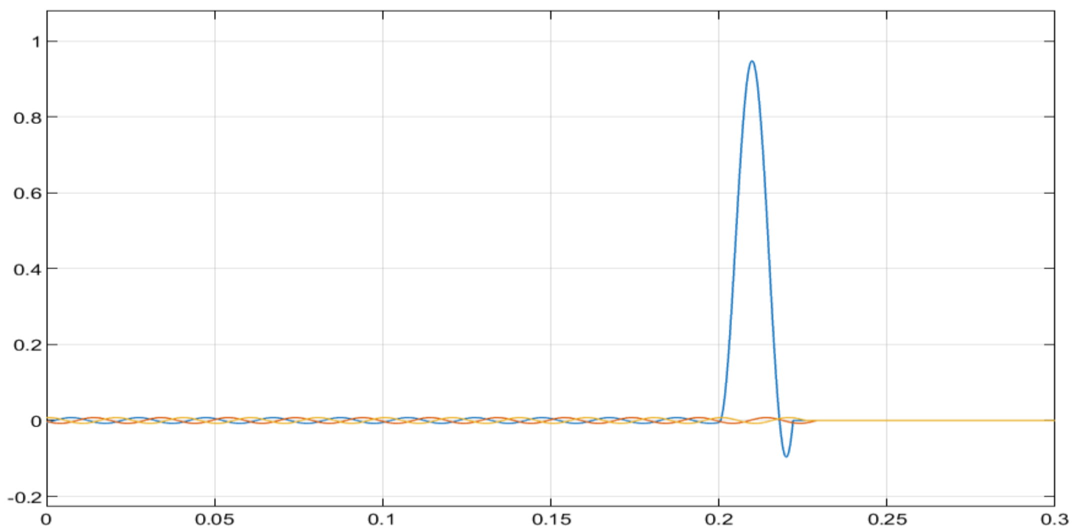



Figure 24. Primary-Side Current $I_1$ during Single-Phase-to-Ground Internal Fault.

Figure 25. shows the secondary-side current $I_2$ during the single-phase-to-ground internal fault. The current waveform becomes disturbed after the fault is applied. After the relay operates and the circuit breaker opens, the current decreases to nearly zero.

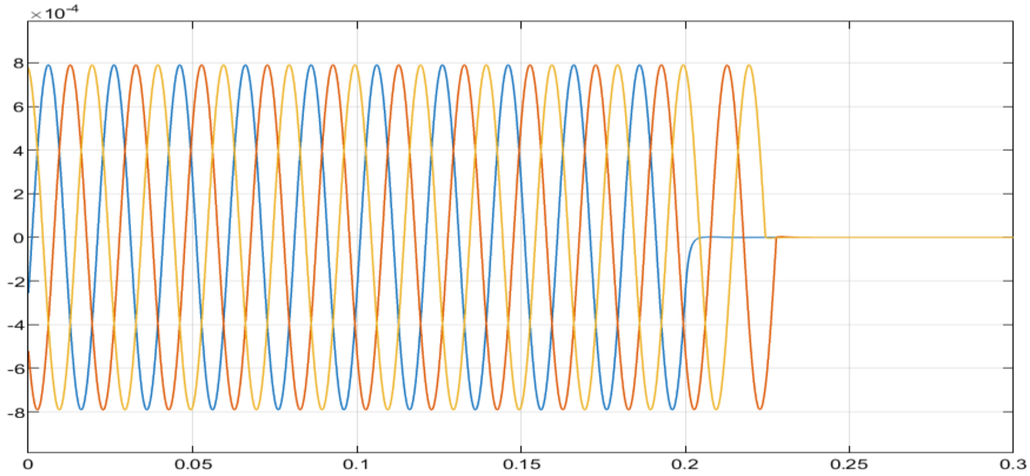



Figure 25. Secondary-Side Current $I_2$ during Single-Phase-to-Ground Internal Fault.

Table A. summarizes the relay response under all simulated scenarios. The results show that the relay operates correctly for internal faults and remains stable during normal operation, external faults, and transformer energization. This confirms that the proposed differential relay can distinguish between fault and non-fault conditions.

| Test Case | Fault Location / Condition | Expected Relay Response | Simulation Result |
|---|---|---|---|
| Normal Operation | No fault | No trip | Trip = 0, CB_cmd = 1 |
| Three-Phase-to-Ground Internal Fault | Inside protection zone | Trip | Trip = 1, CB_cmd = 0 |
| Three-Phase-to-Ground External Fault | Outside protection zone | No Trip | Trip = 0, CB_cmd = 1 |
| Transformer Inrush Current | Transformer energization | No trip | increased, Inrush_Block = 1, Trip = 0, CB_cmd = 1 |
| Single-Phase-to-Ground Internal Fault | Inside protection zone | Trip | Trip = 1, CB_cmd = 0 |

Table A. Summary of Relay Response under Different Test Cases.

## Conclusion

This project presented the design and simulation of a power transformer differential protection relay using MATLAB/Simulink. The relay was tested under different Scenarios including normal operation, internal faults, external faults, transformer energization and single-phase-to-ground faults.

The simulation results showed that the relay successfully detected internal faults and generated the appropriate trip signal to isolate the transformer. At the same time the relay remained stable during external faults and normal operating conditions. The second harmonic blocking method effectively prevented false tripping during transformer energization and magnetizing inrush current conditions.

The implemented relay algorithm demonstrated good selectivity, sensitivity and stability. The proposed differential relay model can provide reliable transformer protection under different power system Scenarios.

## References

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## Appendix. Additional Simulation Results

Figure A1. shows the restraint current ($I_r$) during the internal fault test. The restraint current increases as the through-fault current increases which helps the relay remain stable and avoid unwanted tripping.

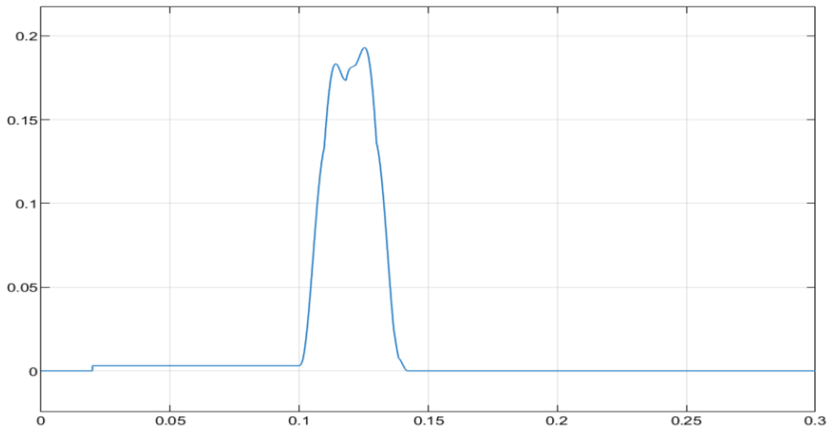


Figure A1. Restraint Current $I_r$ during Internal Fault.

Figure A2. shows the primary-side and secondary-side voltage waveform during the internal fault test. The voltage remains sinusoidal before the fault then becomes disturbed when the internal fault occurs. After the relay trips and the circuit breaker opens the voltage drops to nearly zero.

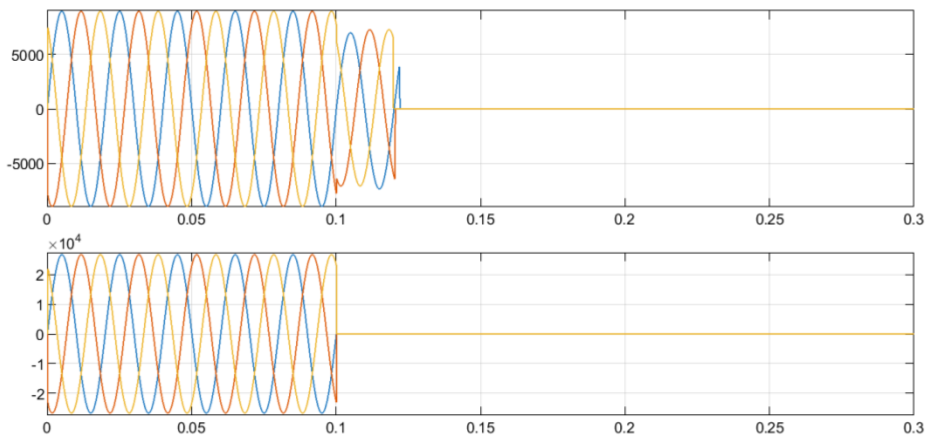


Figure A2. Primary-Side and secondary-side Voltage during Internal Fault.